Year 2 Assessed Problems

Semester 1

Assessed Problems 9

SOLUTIONS TO BE SUBMITTED ON CANVAS BY

Wednesday 10th December

at 17:00

Quantum Mechanics 2 - Problem Sheet 3

A particle of mass m moves in one-dimension with the potential equal to zero. The particle is in a state described by the following wavefunction, which is a superposition of two energy eigenstates and is normalised to give unit probability over a length a.

$$\Psi(x,t) = \sqrt{\frac{2}{3a}} \left[\cos\left(\frac{\pi x}{a}\right) e^{-iE_1 t/\hbar} + \sqrt{2}\cos\left(\frac{3\pi x}{a}\right) e^{-iE_3 t/\hbar} \right]$$

(a) Express the values E_1 and E_3 in terms of a, m and $\hbar$.
If the energy of the particle is measured, what possible values can be obtained and with what probabilities? [5]

(b) If instead the momentum of the particle is measured, what can be the possible results and what are their probabilities? [3]

(c) If the energy is measured first and yields the higher of the possible values, what results could a subsequent measurement of the momentum yield and with what probabilities? [2]

Year 2 Particles and Nuclei
Problem Set 3

Problem 1 The charged pion π^- is a meson and primarily decays into a final state of a muon, μ^-, and an anti-muon neutrino, $\bar{\nu}_\mu$.

If the π^- decays at rest, what is the momentum of the μ^- and the $\bar{\nu}_\mu$? Assume one-dimension along the z-axis with the μ^- in the positive z-direction and the $\bar{\nu}_\mu$ in the negative z-direction.

What are the energy-momentum vectors of the μ^- and $\bar{\nu}_\mu$ in a frame of reference where the π^- was travelling along the positive z-direction with a velocity of $0.6c$?

[5 points]

Quantum Approach to Solids - Problem Sheet 3

Using a numerical integration method of your choice, calculate the Debye heat capacity for a monatomic solid over a temperature range from 0 K to 300 K, using Debye temperatures of 50 K and 750 K.

Plot both of your results on the same graph, making sure that the axes are labelled, and that there is a legend. Your temperature axis should stop at 300 K. Include a dashed line across the full temperature range indicating the Dulong-Petit value of the heat capacity. **[3]**

Next, consider your calculated heat capacities over the smaller temperature range of 0 to 50 K. Re-plot the data on three separate graphs, this time with axes of C/T vs T^2. Again, your axes and curves should be appropriately labelled.
Comment briefly on the curves obtained. **[2]**

As a reminder, the Debye heat capacity is given by:

$$C_V = 9Nk_B(\frac{T}{\Theta_D})^3 \int_0^{\frac{\Theta_D}{T}} \frac{x^4 \mathrm{e}^x}{(\mathrm{e}^x - 1)^2} \mathrm{d}x.$$

Your work will only be assessed on the basis of the output obtained. However, you must include a copy of the code used, or other proof of method used, with your submission.

5 Non-cartesian vector fields

★*Problem 5.1* **Line integrals**

(i)

$$\mathbf{F} = x^2 y\,\hat{\mathbf{x}} + xz\,\hat{\mathbf{y}} - 2yz\,\hat{\mathbf{z}}$$

and Γ is the curve whose parametric equation is

$$\mathbf{r}(s) = \hat{\mathbf{x}}\,4s + \hat{\mathbf{y}}\,2s^2 + \hat{\mathbf{z}}\,s^3\ .$$

Evaluate the line integral

$$\int_{\mathbf{r}_1}^{\mathbf{r}_2} \mathbf{F}\cdot \mathrm{d}\mathbf{r}$$

along the path $\mathbf{r}(s)$ which joins the points $\mathbf{r}_1 = (0,0,0)$ and $\mathbf{r}_2 = (4,2,1)$. **[3]**

(ii) In spherical polar coordinates (r,θ,φ),

$$\mathbf{F} = \hat{\mathbf{r}}\,r^2 + \hat{\boldsymbol{\theta}}\,r\mathrm{e}^{2\theta} + \hat{\boldsymbol{\varphi}}\,r^3\cos\varphi\ .$$

Calculate $\nabla\cdot\mathbf{F}$. **[3]**

(iii) In spherical polar coordinates (r,θ,φ),

$$\mathbf{F} = \hat{\mathbf{r}}\,r^2 + \hat{\boldsymbol{\theta}}\,r\,\mathrm{e}^{2\theta} + \hat{\boldsymbol{\varphi}}\,r^3\cos\varphi\ ,$$

Calculate $\nabla\times\mathbf{F}$. **[4]**